**Aaron M. Bornstein** **Updated: October 19, 2026**

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CURRENT POSITION

2019- Assistant Professor, Department of Cognitive Sciences
Faculty, Institute for Mathematical Behavioral Sciences
Fellow, Center for the Neurobiology of Learning and Memory
University of California, Irvine

PREVIOUS APPOINTMENTS

2019-2025 Assistant Professor, Department of Cognitive Sciences, UC Irvine
2018 Associate Research Scholar, Princeton Neuroscience Institute
2013-2018 Postdoctoral Research Associate, Princeton Neuroscience Institute
2007-2013 Graduate researcher, New York University
2006 Research assistant, Deicken lab, UCSF/VA
2005-2007 Research assistant, Wagner lab, Stanford

EDUCATION

2013 Ph.D., Cognition & Perception, **New York University**
2003 S.B., Mathematics (additional concentration in Economics)
**Massachusetts Institute of Technology**

PUBLICATIONS

(∗ = *Equal contribution*; + = *UCI Lab member.*)

*Preprints*

PP7 Copeland A, Stafford T, Crook C, Pun M, **Bornstein AM**, Hogarth L, Murphy JG, Pickard H, Ryan F, Strickland JC, Field M. The effect of alcohol-specific episodic memories and alcohol cue exposure on value-based decision-making and ad libitum alcohol consumption. *PsyArXiv*.
doi:10.31234/osf.io/3ka57

PP6 Khoudary A+, Peters MAK*, **Bornstein AM***. Memory retrieval explains dynamic effects of expectations on perceptual decisions. *bioRxiv*.
doi:10.64898/2026.09.22.753289

PP5 Leonard BT+, Martínez-Ortíz MA, Bock J, Zhang Y+, Flores S, Taylor DV, Glynn LM, Davis EP, Stern HS, Baram TZ, Hartley CA, Yassa MA*, **Bornstein AM***. Anhedonia counteracts the effects of early-life unpredictability on threat-reward decision-making. *bioRxiv*.
doi:10.64898/2026.05.16.725643

PP4 Wang W, Chierchia G, Cooley BJ, Chang T, **Bornstein AM**, Schweizer S. Autobiographical Memory Specificity Shapes Navigation of Social Uncertainty in Depression. *PsyArXiv*.
doi:10.31234/osf.io/z6vej

PP3 Yoo J+, **Bornstein AM**. Temporal dynamics of model-based control reveal arbitration between multiple task representations. *PsyArXiv*.
doi:10.31234/osf.io/sgcy5

PP2 Zhang Y+*, Chen Y+*, Chen XR+, Harhen NC+, Glynn LA, Davis EP, Baram TZ, Risbrough VB, Stout DA, **Bornstein AM**. A specific computational role for early-life, and not lifelong, stress in decision-making under uncertainty. *bioRxiv*.
doi:10.64898/2026.06.01.729407

PP1 Zhou D+, Noh SM+, Harhen NC+, Banavar NV+, Kirwan B, Yassa MA, **Bornstein AM**. A compressed code for memory discrimination. *bioRxiv*.
doi:10.1101/2025.10.12.681901

*Peer-reviewed journal articles*

JP28 Harhen NC+, **Bornstein AM***, Hartley CA* (2026). Developmental changes in memory structure and precision alter the use of retrieved episodes during decisions for reward. *Proceedings of the National Academy of Sciences*, 123(31):e2525494123.
doi:10.1073/pnas.2525494123

JP27 Harhen NC+, Budiono R, Hartley CA*, **Bornstein AM*** (2026). Structure inference in complex environments improves from childhood to adulthood. *Developmental Science*, 29(3):e70163.
doi:10.1111/desc.70163

JP26 Noh SM+*, Zhou D+*, Cooper KW, Guo S+, Dinh ET+, **Bornstein AM** (2026). Sparsity and memory constraints interact with training sequence to bias learning of associative maps. *Neuropsychologia*, 225:109396.
doi:10.1016/j.neuropsychologia.2026.109396

JP25 Noh SM+*, Cooper KW*, Guo S+, Zhou D+, Stark CEL, **Bornstein AM** (2026). Multi-step inference can be improved across the human lifespan with individualized memory interventions. *The Journals of Gerontology, Series B: Psychological Sciences and Social Sciences*, 81(5):gbag038.
doi:10.1093/geronb/gbag038

JP24 Hadj-Amar B, **Bornstein AM**, Guindani M, Vannuci M (2025). Discrete Autoregressive Switching Processes with Cumulative Shrinkage Priors for Graphical Modeling of Time Series Data. *Journal of Computational and Graphical Statistics*.
doi:10.1080/10618600.2025.2551271

JP23 Schetzsle B, Lee J, **Bornstein AM**, Shahbaba B, Guindani M (2025). A Bayesian Time-Varying Psychophysiological Interaction Model. *Data Science in Science*, 4(1):2519436.
doi:10.1080/26941899.2025.2519436

JP22 Khoudary A+, Peters MAK*, **Bornstein AM*** (2025). Reasoning goals and representational decisions in computational cognitive neuroscience: lessons from the drift diffusion model. *European Journal of Neuroscience*, 61:e7009.
doi:10.1111/ejn.70098

JP21 Banavar NV+, Noh SM+, Wahlheim CN, Cassidy BS, Kirwan CB, Stark CEL, **Bornstein AM** (2024). A response time model of the three-choice Mnemonic Similarity Task

provides stable, mechanistically interpretable individual-difference measures. *Frontiers in Human Neuroscience*, 18:137928.
doi:10.3389/fnhum.2024.137928

JP20 Yoo J+, Chrastil ER, **Bornstein AM** (2024). Cognitive graphs: Representational substrates for planning. *Decision*, 11(4), 537-556.
doi:10.1037/dec0000249

JP19 Chen J, **Bornstein AM** (2024). The causal structure and computational value of narratives. *Trends in Cognitive Sciences*, 28(8):769-781.
doi:10.1016/j.tics.2024.04.003

JP18 Harhen NC+, **Bornstein AM** (2024). Interval timing as a computational pathway from early life adversity to affective disorders. *Topics in Cognitive Science*, 16(2024):92-112.
doi:10.1111/tops.12701

JP17 Noh SM+, Singla UK, Bennett IJ, **Bornstein AM** (2023). Memory precision and age differentially predict the use of decision-making strategies across the lifespan. *Scientific Reports*, 13:17014.
doi:10.1038/s41598-023-44107-5

JP16 **Bornstein AM**, Aly M, Feng SF, Turk-Browne NB, Norman KA, Cohen JD (2023). Associative memory retrieval modulates upcoming perceptual decisions. *Cognitive, Affective, & Behavioral Neuroscience*, 23:645665.
doi:10.3758/s13415-023-01092-6
doi:10.18112/openneuro.ds001614.v1.0.1

JP15 Harhen NC+, **Bornstein AM** (2023). Overharvesting in human patch foraging reflects rational structure learning and adaptive planning. *Proceedings of the National Academy of Sciences*, 120(13):e2216524120.
doi:10.1073/pnas.2216524120

JP14 Otto AR, Devine S, Schultz E, **Bornstein AM***, Louie K* (2022). Context-dependent choice and evaluation in real-world consumer behavior. *Scientific Reports*, 12:17744.
doi:10.1038/s41598-022-22416-5
doi:10.17605/OSF.IO/EC5DX

JP13 Rmus M, Ritz H, Hunter LE, **Bornstein AM***, Shenhav A* (2022). Humans can navigate complex graph structures acquired during latent learning. *Cognition*, 225:105103.
doi:10.1016/j.cognition.2022.105103

JP12 Wang S, Feng SF, **Bornstein AM** (2021). Mixing memory and desire: How decisions for reward depend on the dynamics and content of memory reinstatement. *Wiley Interdisciplinary Reviews: Cognitive Science*, e1581.
doi:10.1002/wcs.1581
**Featured on the cover.**
**Recognized as one of the top-ten most-cited papers from this journal/year.**

JP11 Rouhani N, Norman KA, Niv Y, **Bornstein AM** (2020). Reward prediction errors create event boundaries in memory. *Cognition*, 203:104269.
doi:10.1016/j.cognition.2020.104269

JP10 **Bornstein AM***, Pickard H* (2020). Chasing the first high: Memory sampling in drug choice. *Neuropsychopharmacology*, 45(6):907-915.
doi:10.1038/s41386-019-0594-2 **Featured on the cover.**

JP9 Kane GA, **Bornstein AM**, Shenhav A, Wilson RC, Daw ND, Cohen JD (2019). Rats exhibit similar biases in foraging and intertemporal choice tasks. *eLife*, 8:e48429. doi:10.7554/eLife.48429

JP8 Millner AJ, den Ouden HEM, Gershman SJ, Glenn CR, Kearns J, **Bornstein AM**, Marx BP, Keane TM, Nock MK (2019). Suicidal thoughts and behaviors are associated with an increased decision-making bias for active responses to escape aversive states. *Journal of Abnormal Psychology*, 128(2):106-118. doi:10.1037/abn0000395

JP7 Hoskin AN, **Bornstein AM**, Norman KA, Cohen JD (2018). Refresh my memory: Episodic memory reinstatements intrude on working memory maintenance. *Cognitive, Affective, & Behavioral Neuroscience*, 19:338-354. doi:10.3758/s13415-018-00674-z doi:10.18112/openneuro.ds001576.v1.0.0

JP6 **Bornstein AM**, Khaw MW, Shohamy D, Daw ND (2017). Reminders of past choices bias decisions for reward in humans. *Nature Communications*, 8:15958. doi:10.1038/ncomms15958

JP5 **Bornstein AM**, Norman KA (2017). Reinstated episodic context guides sampling-based decisions for reward. *Nature Neuroscience*, 20:997-1003. doi:10.1038/nn.4573 doi:10.18112/openneuro.ds001607.v1.0.1

JP4 **Bornstein AM**, Daw ND (2013). Cortical and hippocampal correlates of deliberation during model-based decisions for rewards in humans. *PLoS Computational Biology*, 9(12):e1003387. doi:10.1371/journal.pcbi.1003387

JP3 **Bornstein AM**, Daw ND (2012). Dissociating hippocampal and striatal contributions to sequential prediction learning. *European Journal of Neuroscience*, 35:1011-1023. doi:10.1111/j.1460-9568.2011.07920.x

JP2 **Bornstein AM**, Daw ND (2011). Multiplicity of control in the basal ganglia: computational roles of striatal subregions. *Current Opinion in Neurobiology*, 21(3):374-380. doi:10.1016/j.conb.2011.02.009

JP1 Preston AR, **Bornstein AM**, Hutchinson JB, Gaare ME, Glover GH, Wagner AD (2010). High-resolution fMRI of content-sensitive subsequent memory responses in human medial temporal lobe. *Journal of Cognitive Neuroscience*, 22:156-173. doi:10.1162/jocn.2009.21195

## Awards

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|---|---|
| 2024 | CNLM Exceptional Mentor Award |
| 2023 | Election to the Memory Disorders Research Society |
| 2020 | Association for Psychological Science "Rising Star" Award |
| 2020 | Brain and Behavior Research Foundation NARSAD Young Investigator Award |
| 2005,6,8 | Honorable mention, NSF Graduate Research Fellowship |
| 2000 | Runner up, MIT $50k Business Plan competition |

## Teaching/Service

*UC Irvine*

Spring 2027 Neural Data Analysis (Undergraduate)

Fall 2026 Cognitive Maps (Graduate; cross-listed with Critical Theory Emphasis)

Fall 2025 Resisting "Neuro-narratives" (Undergraduate)

Winter 2025,2027 LIFTED UC Irvine Prison Education Program (Human Memory)

Spring 2023-2026,Winter 2027 Human Memory (Undergraduate)

Spring 2023-24,Fall 2025 Decision making (Graduate)

Spring 2021-22 Decision making & Problem solving (Graduate; with Prof. Zygmunt Pizlo)

Spring 2020 Topics in Reinforcement Learning (Graduate; with Prof. Mimi Liljeholm)

Spring 2019-22 Research in Exp Psych (Undergraduate; with Prof. Nadia Chernyak)

Winter 2019-22 Advanced Experimental Psychology (Undergraduate)

*Other teaching/mentorship*

Fall 2020 Neuromatch academy mentor

Summer 2020 Neuromatch academy group leader

Summer 2020 GSMI Cientifico Latino

2019 MEET alumni mentor

Summer 2018 Computational & Cognitive Neuroscience Summer School, Suzhou, China

2016-2018 Princeton Prison Teaching Initiative

Summer 2007,08 MIT Middle East Education through Technology (MEET), Jerusalem

*University/Department service*

2024- UCI Research Imaging Equipment Committee

2022-2025 Cognitive Sciences Colloquium committee, faculty advisor

2021- Irvine Faculty Association, Executive board member; 2023-2025 Treasurer; 2025- Co-Chair

2020 CNLM "Evening to Remember" organizational committee

2019- First-Generation Faculty Initiative

2019- UCI Prison Education Program, advisory board, curriculum committee

2019,23 Cognitive Sciences faculty search committee

2018,20,22,24 Cognitive Sciences PhD admissions committee

*Organizational/Field service*

2025- Associate Editor, *Cognitive Science*.

2025- Justice, Diversity, Equity, and Inclusion Committee, *Memory Disorders Research Society*.

2025- Advisor, *All People's Health Collective*.

2024- Editorial board, *Translational Neuroscience*.

2023 UCI Conte Center annual symposium, organized with the Conte Center Team.

2022 NSF/Simons NeuroDataScience workshop, Co-Organizer (with Norbert Fortin and Babak Shahbaba)

2021 Center for Neurobiology of Learning and Memory Spring Meeting Co-organizer (with Lulu Chen)

2018 Princeton Neuroscience Institute "Inside-Out" seminar series Co-organizer (with Ahmed El Hady)

2018 "Goal-Directed Decision Making: Computations and Circuits" *Elsevier* Co-editor (with Richard Morris & Amitai Shenhav)

2015 COSYNE Workshop "Memory in action: The role(s) of the hippocampus in decisions for reward" Co-organizer (with G. Elliott Wimmer)